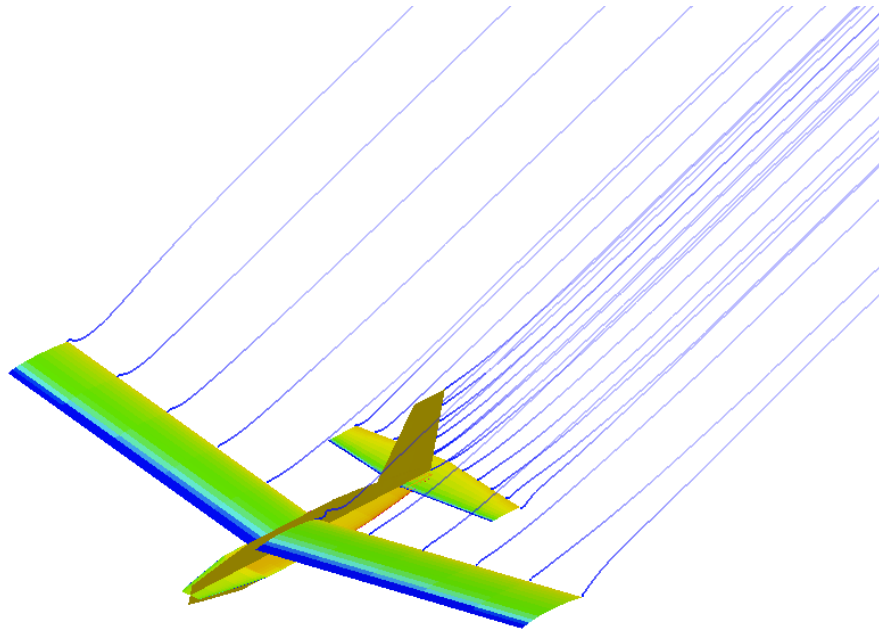


ENAE 410 – Design Lab 2

Using OpenVSP and Lift/Drag Buildup to analyze an aircraft



1 Objective

The primary objective of this exercise is to familiarize students with the hands-on application of OpenVSP and the Lift/Drag buildup approach in analyzing aerodynamic coefficients of a specific aircraft geometry. Students will compare their results with provided experimental data, recognizing differences, and encouraging meaningful discussions.

2 Background

2.1 Motivation

In modern aerodynamic design, software tools are indispensable for the simulation and evaluation of aircraft performance. Through this activity, students will gain direct experience with these tools, significantly enriching their understanding of the principles and techniques employed in the field of aerodynamics. The discrepancies between simulated and experimental results further highlight the intricacies of real-world aerodynamics, prompting critical thinking and discussion.

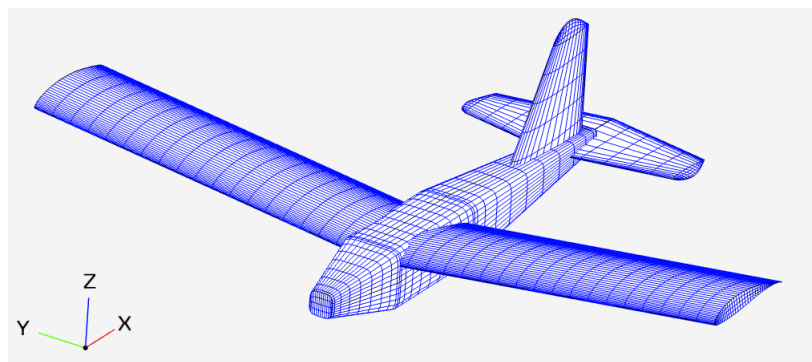
Software tools like OpenVSP and analytical methods such as the Lift/Drag buildup approach are central to understanding and predicting aircraft performance in modern aerodynamics. This activity gives students the opportunity to apply theoretical knowledge in a practical setting, while also promoting critical thinking about the complexities of real-world aerodynamics.

2.2 Overview of OpenVSP

Open Vehicle Sketch Pad (OpenVSP) is an open-source software developed by NASA that allows engineers to construct full aircraft configurations in a 3D parametric modeling environment. The tool is used to generate three-dimensional models of aircraft that are suitable for aerodynamic analysis, stability analysis, performance analysis, and more. OpenVSP's user-friendly interface, accessibility, and powerful features make it an excellent resource for students to learn essential aerodynamics modeling skills.

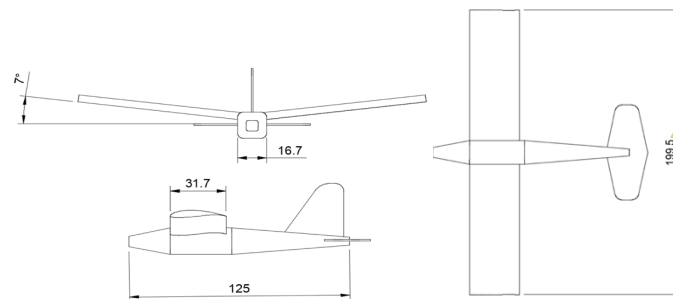
3 Geometry Definition

In this activity, students will analyze a predefined aircraft geometry that has been provided. This model represents a small, yet intricately designed aircraft with a conventional wing-body-tail configuration common in aviation.



The aircraft wing has a Wortmann FX 63-137 airfoil, an airfoil renowned for its high lift-to-drag ratio at low Reynolds numbers, making it an ideal choice for small, low-speed aircraft. The airfoil's unique shape, characterized by a flat bottom and a curved top, enables the aircraft to maintain sufficient lift during low-speed flights while keeping the drag to a minimum.

An accompanying figure provides a detailed visualization of the aircraft's geometry, giving students a comprehensive understanding of its three-dimensional configuration. Furthermore, an additional figure presents the relevant dimensions of the model, such as wingspan, wing area, fuselage length, tail area, etc. These dimensions are essential for performing both the Lift/Drag buildup and the OpenVSP analyses.



Finally, a table of key parameters and reference areas essential for the analysis has been included. These parameters include geometric details and other crucial data such as the reference wing area, mean aerodynamic chord, reference length, and span. Understanding these parameters is pivotal in correctly applying the Lift/Drag buildup approach and OpenVSP to analyze the aircraft's aerodynamics.

Primary Wing	
Wing Area	6334.0 mm ²
Span	200 mm
Chord length	31.67 mm
Dihedral	7 degrees
Horizontal Tail	
Wing Area	1248.71 mm ²
Span	60.8 mm
Airfoil	Flat plate (t/c = 0.1)
Wetted area	Swet = 2456.4 mm ²
Vertical Tail	
Wing Area	852.45 mm ²
Span	31.9 mm
Airfoil	Flat plate (t/c = 0.1)
Wetted area	Swet = 1497 mm ²
Fuselage	
Wetted area	Swet = 5879 mm ²
Length	125 mm
Equivalent diameter (for l/d)	18.45 mm

4 Calculation instructions

4.1 Lift and Drag Build-up

The component build-up method is a widely used method of analysis for estimating aerodynamic performance and is primarily used in the early stage of design and analysis due to its simplicity. This approach estimates the lift and drag curves for the aircraft by individually estimating the lift and drag contributions of each component and adding them together.

For the drag buildup method, we consider the drag to simply be the sum of each component.

$$C_{d,total} = (C_{d,Wing} + C_{d,fuselage} + C_{d,Horizontal\ Stabilizer} + C_{d,Vertical\ Stabilizer}) \times 1.04$$

The overall drag is computed based on:

$$D = 0.5\rho V^2 S_{ref} C_{d,total}$$

Note that in order for this to work, all of the component coefficients must be normalized by S_{ref} . The reference area for an airplane is taken as the projected area of the main wing. **The factor 1.04 is the interference drag factor which simply estimates interference drag as a 4% increase in drag.**

Main Wing Lift

It is customary to only include the lift of the main wing in the aircraft analysis as lift on the tail surfaces changes due to stability requirements (as you will see later in the course). You should note this as a discrepancy between the buildup approach and the wind-tunnel results and OpenVSP analysis.

For the wings you should compute the overall lift coefficient using the lift curve slope given by:

$$a = \frac{a_o}{1 + \frac{a_o}{\pi AR}}$$

Where a_o is the sectional lift coefficient slope ($\frac{dc_l}{d\alpha}$ in 1/rad) of the airfoil (obtained from the data provided) and AR is the wing aspect ratio. Then the lift can be computed using:

$$C_L = a(\alpha - \alpha_{c_l=0})$$

Where $\alpha_{c_l=0}$ is the angle of attack with zero lift. These angles should be in radians. Note: this only works in the region of flight where the lift curve is linear. Therefore, you must manually limit high angle of attacks and stall, and understand that this is a limit of the method used.

Main Wing Drag

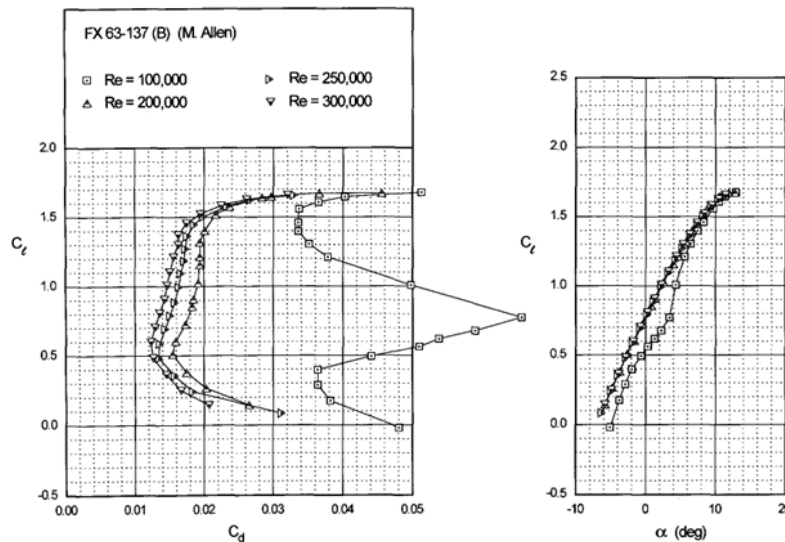
The drag coefficient may be taken as the sum of **profile drag** and **lift induced drag**:

$$C_D = c_d + C_{D,i}$$

where c_d is the profile drag coefficient and $C_{D,i}$ is the induced drag coefficient computed from:

$$C_{D,i} = \frac{C_L^2}{\pi e AR}$$

Here e is the Oswald efficiency factor. For our purposes, assume that it is 0.9. Note that C_L is the overall lift coefficient. The form drag factor can be estimated from the drag polar:



Other component drags

For non-lifting surfaces (including tail plane surfaces), drag is often estimated using a simple model based on “form factor”. This is given by:

$$C_D = \frac{S_{wet} C_f FF}{S_{ref}}$$

There are MANY models for both form factor and C_f (coefficient of friction). Some simple ones are provided here. Note that C_f needs to be recalculated for *each surface* based on the Reynolds number of that component. Also note, S_{ref} should be the main wing area S_w , this allows you to add the drags coefficients together (they all need the same reference area).

The Reynolds number you should use should match the one used in lab 1 (60,000). You should choose the airfoil data closest to this value in the data provided, or you may search for more data online. Note this Reynolds number is based on the wing chord length, and is therefore only the Reynolds number of the wing. In the buildup approach you should compute the Reynolds number of each component based on length.

Friction factor

Laminar ($Re < 250,000$)	Turbulent ($Re > 250,000$)
$C_f = \frac{1.328}{\sqrt{Re}}$	$C_f = \frac{0.455}{(\log_{10} Re)^{2.58}}$

Fuselage

Fuselage drag form factor can be estimated using the following equation where FR is the “fineness ratio” [3]:

$$FF = 1 + \frac{60}{FR^3} + 0.0025FR$$

Where the fineness ratio is given by the ratio between the body length, l , and the diameter, d :

$$FR = l/d$$

Horizontal and Vertical Stabilizers

The form factor for the stabilizers can be taken as [3]:

$$FF = 1 + 2\left(\frac{t}{c}\right) + 60\left(\frac{t}{c}\right)^4$$

where t/c is the thickness to chord ratio of the wing.

4.2 OpenVSP

The following shows the parameters you need to input in order to create an OpenVSP model for the provided aircraft. The geometry is split up into the main components: The main wing, fuselage, and stabilizers. Each must be created within the OpenVSP software in order to be analyzed. The following sections illustrate how to define the geometry, and, subsequently how to analyze it using VSPAero.

4.2.1 Geometry Definition

The following screenshots show what to input in the software.

Fuselage parameters:

Cross Section

<< < 0 > >>

Insert Cut Copy Paste

Num U: 6

Fwd. Cluster: 1.00000

Aft Cluster: 1.00000

X: 0.00000 0.00000

Y: 0.00000 0.00000

Z: 0.00000 0.00000

Rot X: 0.00000

Rot Y: 0.00000

Rot Z: 0.00000

Spin: 0.00000

Type

Choose Type: SUPER_ELLIPSE Show Convert CEDIT

☒ Width: 6.60000

☐ Area: 41.98248

☒ Height: 6.60000

☐ H/W Ratio: 1.00000

MaxWLoc: 0.00000

M: 6.00000

N: 6.00000

T/B Symmetric Exponents

M Bottom: 6.00000

N Bottom: 6.00000

Cross Section

<< < 1 > >>

Insert Cut Copy Paste

Num U: 10

Fwd. Cluster: 1.00000

Aft Cluster: 1.00000

X: 0.18640 23.30000

Y: 0.00000 0.00000

Z: 0.00000 0.00000

Rot X: 0.00000

Rot Y: 0.00000

Rot Z: 0.00000

Spin: 0.00000

Type

Choose Type: SUPER_ELLIPSE Show Convert CEDIT

☒ Width: 16.67000

☐ Area: 267.82518

☒ Height: 16.67000

☐ H/W Ratio: 1.00000

MaxWLoc: 0.00000

M: 6.00000

N: 6.00000

T/B Symmetric Exponents

M Bottom: 6.00000

N Bottom: 6.00000

Cross Section

<< < 2 > >>

Insert Cut Copy Paste

Num U: 12

Fwd. Cluster: 1.00000

Aft Cluster: 1.00000

X: 0.46640 58.30000

Y: 0.00000 0.00000

Z: 0.00000 0.00000

Rot X: 0.00000

Rot Y: 0.00000

Rot Z: 0.00000

Spin: 0.00000

Type

Choose Type: SUPER_ELLIPSE Show Convert CEDIT

☒ Width: 16.66700

☐ Area: 267.72879

☒ Height: 16.66700

☐ H/W Ratio: 1.00000

MaxWLoc: 0.00000

M: 6.00000

N: 6.00000

T/B Symmetric Exponents

M Bottom: 6.00000

N Bottom: 6.00000

Cross Section

<< < 3 > >>

Insert Cut Copy Paste

Num U: 12

Fwd. Cluster: 1.00000

Aft Cluster: 1.00000

X: 1.00000 125.00000

Y: 0.00000 0.00000

Z: 0.00120 0.15000

Rot X: 0.00000

Rot Y: 0.00000

Rot Z: 0.00000

Spin: 0.00000

Type

Choose Type: SUPER_ELLIPSE Show Convert CEDIT

☒ Width: 5.00000

☐ Area: 24.09463

☒ Height: 5.00000

☐ H/W Ratio: 1.00000

MaxWLoc: 0.00000

M: 6.00000

N: 6.00000

T/B Symmetric Exponents

M Bottom: 6.00000

N Bottom: 6.00000

Skin Cross Section

<< 0 >>

Top Side All Sym

Enforce	C0	Set	=	Set	
☒	0.00	☒	Angle	☒	0.00
☒	0.00	☒	Slew	☒	0.00
☒	0.00	☒	Strength	☒	0.00
☒	6.00	☐	Curvature	☐	6.00

Right Side

Enforce	C0	Set	=	Set	
☒	0.00	☒	Angle	☒	0.00
☒	0.00	☒	Slew	☒	0.00
☒	0.00	☒	Strength	☒	0.00
☒	6.00	☐	Curvature	☐	6.00

Bottom Side T/B Sym

Enforce	C0	Set	=	Set	
☒	0.00	☒	Angle	☒	0.00
☒	0.00	☒	Slew	☒	0.00
☒	0.00	☒	Strength	☒	0.00
☒	6.00	☐	Curvature	☐	6.00

Left Side R/L Sym

Enforce	C0	Set	=	Set	
☒	0.00	☒	Angle	☒	0.00
☒	0.00	☒	Slew	☒	0.00
☒	0.00	☒	Strength	☒	0.00
☒	6.00	☐	Curvature	☐	6.00

Design

Length 125.0000

Design Policy

XSec Order: MONOTONIC

Tip Treatment

Nose Cap Type Flat

Length 1.00000
Offset 0.00000
Strength 0.50000

Sweep Stretch

Tail Cap Type Flat

Length 1.00000
Offset 0.00000
Strength 0.50000

Sweep Stretch

Cap Tess 3

Engine Definition

Geom Type: NONE
Inlet Type: FLOWTHROUGH
Outlet Type: OUTLET TO LIP

Inlet Lip: Indx 0 U 0.00000
Inlet Face: Indx 0 U 0.00000
Outlet Face: Indx 0 U 0.00000
Outlet Lip: Indx 0 U 0.00000

Engine Representation

Inlet Mode: TO LIP
Outlet Mode: TO LIP

Wing parameters

Gen XForm Sub Plan Sect Airfoil Blending Modify

Transforms

Coord System: ☐ Rel ☒ Abs

XLoc:

YLoc:

ZLoc:

XRot:

YRot:

ZRot:

Rot Origin(X):

Symmetry

About: ☒ 1 MainWing ☐ Attach ☐ Object

Planar: ☐ XY ☒ XZ ☐ YZ

Axial: ☒ None ☐ X ☐ Y ☐ Z

N:

Scale Factor

Scale:

Gen XForm Sub Plan Sect Airfoil Blending Modify

Airfoil Section

Type

Choose Type:

Read File

Name:

Chord:

T/C:

Base T/C:

Gen XForm Sub Plan Sect Airfoil Blending Modify

Total Planform

Span:

Proj Span:

Chord:

Area:

Aspect Ratio:

Tip Treatment

Cap Tess:

Root Cap Type:

Length:

Offset:

Strength:

Sweep Stretch

Tip Cap Type:

Length:

Offset:

Strength:

Root Incidence

Incidence:

Incidence Loc:

Tessellation Control

LE Clustering:

TE Clustering:

Minimum LE/TE Panel Width:

Maximum Growth Ratio:

Wing Loft Control

Twist Reference: ☐ Rel ☒ Abs

Dihedral Reference: ☐ Rel ☒ Abs

☒ Rotate All Foils To Match Dihedral

☐ Rotate Wing Root Foil To Match Dihedral

☒ Correct Foil Thickness For Dihedral Rotation

Gen XForm Sub Plan Sect Airfoil Blending Modify

Wing Section

Num Sections:

Interpolated XSecs

Num U:

RL Cluster:

Tip Cluster:

Section Planform

☐ AR ☐ Span ☐ Area ☐ Taper ☐ Ave C ☒ Root C ☒ Tip C ☐ Sec SW

Projected Span:

Sweep

Sweep:

Sweep Loc:

Sec SW Loc:

Twist

Twist:

Twist Loc:

Dihedral

Dihedral:

Horizontal tail parameters

Gen XForm Sub Plan Sect Airfoil Blending Modify

Total Planform

Span

Proj Span

Chord

Area

Aspect Ratio

60.81133

60.81133

20.53417

248.70998

2.96147

Tip Treatment

Cap Tess

Root Cap Type

Length

Offset

Strength

5

Flat

1.00000

0.00000

0.50000

Sweep Stretch

Tip Cap Type

Length

Offset

Strength

Round

4.00000

0.00000

1.00000

Sweep Stretch

Root Incidence

Incidence

Incidence Loc

0.00000

0.25000

Tessellation Control

LE Clustering

TE Clustering

Minimum LE/TE Panel Width

Maximum Growth Ratio

0.25000

0.25000

0.02556

1.437

Wing Loft Control

Twist Reference:

Dihedral Reference:

Rotate All Foils To Match Dihedral

Rotate Wing Root Foil To Match Dihedral

Correct Foil Thickness For Dihedral Rotation

Rel

Abs

Rel

Abs

Gen XForm Sub Plan Sect Airfoil Blending Modify

Wing Section

<<

<

1

>

>>

Split

Cut

Copy

Paste

Insert

Num Sections

1

Interpolated XSecs

Num U

Rt. Cluster

Tip Cluster

6

1.00000

1.00000

Section Planform

AR

Span

Area

Taper

Ave C

Root C

Tip C

Sec SW

Projected Span

1.48074

30.40566

624.35499

0.53507

20.53417

26.75333

14.31500

-11.11824

30.4057

Sweep

Sweep

Sweep Loc

Sec SW Loc

12.00000

0.00000

1.00000

Twist

Twist

Twist Loc

0.00000

0.25000

Dihedral

Dihedral

Rotate Section Tip Foil To Match Dihedral

0.00000

Gen XForm Sub Plan Sect Airfoil Blending Modify

Airfoil Section

<<

<

0

>

>>

Copy

Paste

Type

Choose Type:

ELLIPSE

Show

Convert CEDIT

Height

Width

1.40000

26.75333

Gen XForm Sub Plan Sect Airfoil Blending Modify

Airfoil Section

<<

<

1

>

>>

Copy

Paste

Type

Choose Type:

ELLIPSE

Show

Convert CEDIT

Height

Width

1.40000

14.31500

Gen XForm Sub Plan Sect Airfoil Blending Modify

Transforms

Coord System:

Rel

Abs

XLoc

YLoc

ZLoc

XRot

YRot

ZRot

Rot Origin(X)

110.000

0.000

0.652

0.000

0.000

0.000

0.000

Symmetry

About:

1 TailPlane

Attach

Object

Planar:

XY

XZ

YZ

Axial:

None

X

Y

Z

N

2

Scale Factor

Scale

1.0000

Reset

Accept

Attach To Parent

Translate:

None

Comp

UW

RST

LMN

Rotate:

None

Comp

UW

RST

LMN

Vertical tail parameters

Gen XForm Sub Plan Sect Airfoil Blending Modify

Total Planform

Span	31.98588
Proj Span	31.98588
Chord	26.65083
Area	852.45026
Aspect Ratio	1.20018

Tip Treatment

Cap Tess 6

Root Cap Type None

Length	1.00000
Offset	0.00000
Strength	0.50000

Sweep Stretch

Tip Cap Type Round

Length	6.00000
Offset	0.00000
Strength	1.00000

Root Incidence

Incidence	0.00000
Incidence Loc	0.25000

Tessellation Control

LE Clustering	0.25000
TE Clustering	0.25000

Minimum LE/TE Panel Width 0.05308
Maximum Growth Ratio 1.437

Wing Loft Control

Twist Reference: Rel Abs
Dihedral Reference: Rel Abs

Rotate All Foils To Match Dihedral
Rotate Wing Root Foil To Match Dihedral
Correct Foil Thickness For Dihedral Rotation

Gen XForm Sub Plan Sect Airfoil Blending Modify

Wing Section

1

Split Cut Copy Paste Insert

Num Sections 1

Interpolated XSecs

Num U	6
Rt. Cluster	1.00000
Tip Cluster	1.00000

Section Planform

Span	31.98588
Area	852.45026
Taper	0.44052
Ave C	26.65083
Root C	37.00167
Tip C	16.30000
Sec SW	0.13221

Projected Span 31.9859

Sweep

Sweep	32.81818
Sweep Loc	0.00000
Sec SW Loc	1.00000

Twist

Twist	0.00000
Twist Loc	0.25000

Dihedral

Dihedral	0.00000
----------	---------

Rotate Section Tip Foil To Match Dihedral

Gen XForm Sub Plan Sect Airfoil Blending Modify

Transforms

Coord System: Rel Abs

XLoc	84.887	84.887
YLoc	0.000	0.000
ZLoc	0.000	0.000
XRot	90.000	90.000
YRot	0.000	0.000
ZRot	0.000	0.000

Rot Origin(X) 0.000

Symmetry

About: 1 VerticalStabilizer Attach Object

Planar: XY XZ YZ

Axial: None X Y Z

N 2

Scale Factor

Scale 1.0000 Reset Accept

Attach To Parent

Translate: None Comp UW RST LMN

Rotate: None Comp UW RST LMN

Gen XForm Sub Plan Sect Airfoil Blending Modify

Airfoil Section

0

Copy Paste

Type

Choose Type: ELLIPSE Show Convert CEDIT

Height	1.60000
Width	37.00167

Gen XForm Sub Plan Sect Airfoil Blending Modify

Airfoil Section

1

Copy Paste

Type

Choose Type: ELLIPSE Show Convert CEDIT

Height	1.60000
Width	16.30000

4.2.2 VSPAero Analysis

To analyze the aerodynamics of the aircraft, we will use the VSPAero module. This model uses a method known as a “Vortex Lattice” method. The inner workings of a calculation like this will be covered in later aerodynamics courses. Follow along with the instructor-created video on D2L, which illustrates how to create the geometry and conduct the analysis.

5 Discussion of results

Compare the results of the VSPAero analysis and the component build up method for lift and drag to the experimental data from Lab 1. Think critically about any differences in results. What uncertainty will there be when translating any of these results to the real world? What are the implications of uncertainty in performance on the design?

6 Summary and Expected Outcomes

By the end of this activity, students should have a functional understanding of how to utilize OpenVSP for aerodynamic analysis. Students should be able to interpret aerodynamic coefficients, compare simulated results with experimental data, and engage in productive discussions about any discrepancies and potential causes.

Please perform the calculations described above and summarize in a brief report using the provided template.

7 References and Supporting Materials

[1] OpenVSP official website: <http://www.openvsp.org/>

[2] OpenVSP User Manual: http://www.openvsp.org/wiki/doku.php?id=user_manual

[3] Drag buildup models: <https://openvsp.org/wiki/doku.php?id=parasitedrag>